

Exponentiation

Let  $G$  be a group and let  $g \in G$ . For an integer  $n > 0$  we denote:

- $g^n = \underbrace{g \cdot g \cdot \dots \cdot g}_{n \text{ times}}$  (additive notation:  $ng = \underbrace{g + g + \dots + g}_{n \text{ times}}$ )
- $g^{-n} = \underbrace{g^{-1} \cdot g^{-1} \cdot \dots \cdot g^{-1}}_{n \text{ times}}$  (additive notation:  $(-n)g = \underbrace{-g - g - \dots - g}_{n \text{ times}}$ )
- $g^0 = e$  (additive notation:  $0g = 0$ )

Properties of exponentiation

- $g^{m+n} = g^m \cdot g^n$  (additive notation:  $(m+n)g = (mg) + (ng)$ )
- $g^{mn} = (g^m)^n$  (additive notation:  $(mn)g = m(ng)$ )

**Definition 6.1**

Let  $G$  be a group. An *order* of an element  $g \in G$  is the smallest integer  $n \geq 1$  such that  $g^n = e$ . We write:  $|g| = n$ .

If  $g^n \neq e$  for all  $n \geq 1$  then we say that  $g$  is an element of an *infinite order* and we write  $|g| = \infty$ .

**Note.** If  $g^n = e$  then  $g^{-1} = g^{n-1}$ . In particular, if  $g^2 = e$  then  $g = g^{-1}$ .

**Exercise 6.2.** Recall that the multiplication table of the dihedral group  $D_4$  is as follows:

| $\circ$   | $I$       | $R_{90}$  | $R_{180}$ | $R_{270}$ | $H$       | $V$       | $D$       | $D'$      |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| $I$       | $I$       | $R_{90}$  | $R_{180}$ | $R_{270}$ | $H$       | $V$       | $D$       | $D'$      |
| $R_{90}$  | $R_{90}$  | $R_{180}$ | $R_{270}$ | $I$       | $D'$      | $D$       | $H$       | $V$       |
| $R_{180}$ | $R_{180}$ | $R_{270}$ | $I$       | $R_{90}$  | $V$       | $H$       | $D'$      | $D$       |
| $R_{270}$ | $R_{270}$ | $I$       | $R_{90}$  | $R_{180}$ | $D$       | $D'$      | $V$       | $H$       |
| $H$       | $H$       | $D$       | $V$       | $D'$      | $I$       | $R_{180}$ | $R_{90}$  | $R_{270}$ |
| $V$       | $V$       | $D'$      | $H$       | $D$       | $R_{180}$ | $I$       | $R_{270}$ | $R_{90}$  |
| $D$       | $D$       | $H$       | $D'$      | $V$       | $R_{270}$ | $R_{90}$  | $I$       | $R_{180}$ |
| $D'$      | $D'$      | $V$       | $D$       | $H$       | $R_{90}$  | $R_{270}$ | $R_{180}$ | $I$       |

Find the order of every element of  $D_4$ .

**Exercise 6.3.** Find the order of every element in the group  $\mathbb{Z}_6$ .

### Theorem 6.4

If  $G$  is a finite group and  $g \in G$  then  $|g| < \infty$ .

*Proof.* Consider the sequence

$$g^1, g^2, g^3, \dots \subseteq G$$

Since  $G$  consists of finitely many elements, we must have  $g^m = g^n$  for some  $n > m$ . This gives

$$\begin{aligned} g^{-m}g^m &= g^{-m}g^n \\ e &= g^{n-m} \end{aligned}$$

Thus  $|g| \leq n - m < \infty$ . □

### Theorem 6.5

If  $G$  is a group,  $g \in G$  and  $n \geq 1$  is an integer such that  $g^n = e$ , then  $|g|$  divides  $n$ .

*Proof.* We have

$$n = |g| \cdot q + r$$

for some integers  $q \geq 0$  and  $0 \leq r < |g|$ . We want to show that  $r = 0$ . Assume that it is not true. Then we have

$$e = g^n = g^{|g| \cdot q + r} = g^{|g| \cdot q} \cdot g^r = \left(g^{|g|}\right)^q \cdot g^r = e \cdot g^r = g^r$$

We obtain that  $g^r = e$ . This is however impossible, since  $r < |g|$ . □

### Theorem 6.6

If  $G$  is a group, and  $a, b \in G$  are elements such that  $|a|, |b| < \infty$  and  $ab = ba$  then  $|ab|$  divides  $|a| \cdot |b|$ .

*Proof.* Let  $|a| = m$  and  $|b| = n$ . We have

$$(ab)^{mn} = a^{mn} b^{mn} = (a^m)^n \cdot (b^n)^m = e^n \cdot e^m = e$$

By Theorem 6.5 we get that  $|ab|$  divides  $mn = |a| \cdot |b|$ . □

**Example.** In the dihedral group  $D_4$  take  $a = R_{90}$ ,  $b = R_{180}$ . Then  $a \cdot b = R_{90} \cdot R_{180} = R_{270}$ . We have  $|R_{90}| = 4$ ,  $|R_{180}| = 2$ , so  $|R_{90}| \cdot |R_{180}| = 8$  which is divisible by  $|R_{270}| = 4$ .

**Example.** Theorem 6.6 is not true in general if  $ab \neq ba$ . Take for example  $a, b$  to be two different reflections in the dihedral group  $D_3$ . Then  $|a| = |b| = 2$ , so  $|a| \cdot |b| = 4$ , but  $|ab| = 3$ .

As another example, in the group  $GL(2, \mathbb{R})$  take matrices

$$A = \begin{bmatrix} 0 & 2 \\ \frac{1}{2} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Then  $|A| = 2$  and  $|B| = 2$  but  $|AB| = \infty$ .

### Theorem 6.7

If  $G$  is a group, and  $a \in G$  is element such that  $|a| = n < \infty$  then

$$|a^k| = \frac{n}{\gcd(n, k)}$$

**Exercise 6.8.** Compute the order of the element  $6 \in \mathbb{Z}_{10}$ .

*Proof of Theorem 6.7.* Let  $|a^k| = r$ . Then  $r$  is the smallest positive integer such that  $(a^k)^r = a^{kr} = e$ . Using Theorem 6.5 we obtain that  $r$  is the smallest positive integer such that  $n|kr$ . This means that  $kr$  is the least common multiple of  $n$  and  $k$ :  $kr = \text{lcm}(n, k)$ . This gives:

$$kr = \text{lcm}(n, k) = \frac{nk}{\text{gcd}(n, k)}$$

and so  $r = \frac{n}{\text{gcd}(n, k)}$ .

□